

CE725: Vibration-Based Structural Health Monitoring

Department of Civil Engineering,
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Homework Project 1

Submitted by: Group G7

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1 Question 1: Projectile Trajectory Tracking

1.1 State-Space Modeling

The state vector is defined as

$$X = [u_x, v_x, u_y, v_y]^T.$$

Continuous-Time Model. The system evolves as

$$\dot{X} = A_c X + B_c,$$

where

$$A_c = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -g \end{bmatrix}.$$

The model captures projectile motion under gravity: $\dot{u}_x = v_x$, $\dot{u}_y = v_y$, and $\dot{v}_y = -g$.

Discrete-Time Model. The continuous system was discretized using Zero-Order Hold (ZOH) with sampling time $\Delta t = 0.01$ s (100 fps).

1.2 Q1(a): Displacement Measurements

The measurement vector and matrix are

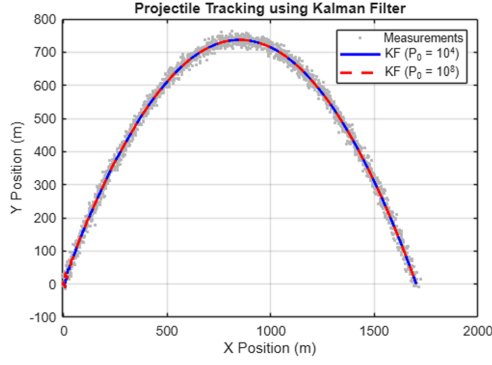
$$z = \begin{bmatrix} u_x \\ u_y \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad R = \text{diag}(100, 100).$$

Kalman filtering was performed with no process noise ($Q = 0$) and two initial covariance cases:

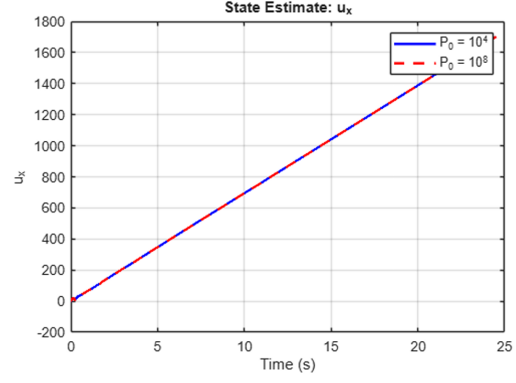
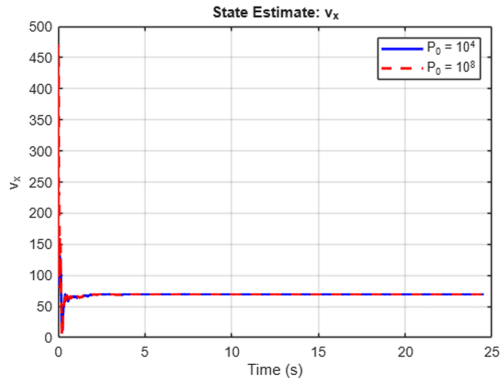
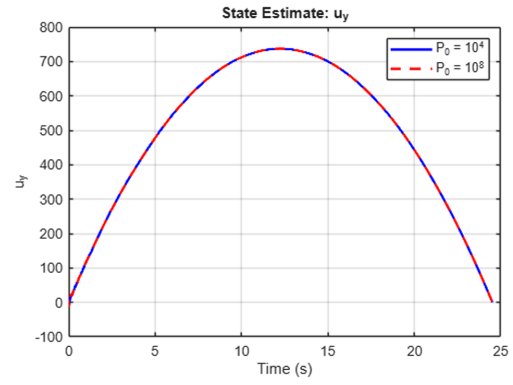
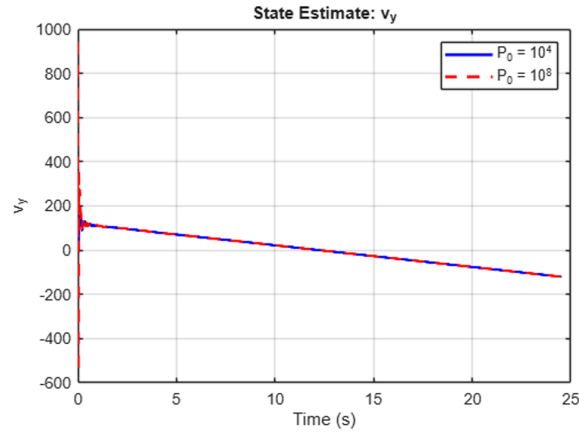
$$P_0 = 10^4 I \quad \text{and} \quad P_0 = 10^8 I.$$

Observations.

- A larger initial covariance ($10^8 I$) results in faster convergence because the filter assumes higher uncertainty in the initial state, allowing it to weight incoming measurements more aggressively.
- A smaller initial covariance ($10^4 I$) produces smoother but slower initial convergence.
- Both cases ultimately converge to nearly identical trajectories, demonstrating the robustness of the Kalman filter to initial condition selection.
- Since displacement states are directly observed, measurement corrections immediately and accurately update position estimates.



(a) Projectile trajectory tracking

(b) State estimate u_x (a) State estimate v_x (b) State estimate u_y Figure 3: State estimate v_y

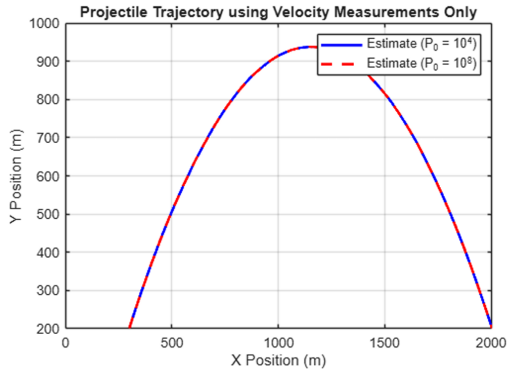
1.3 Q1(b): Velocity Measurements

The measurement vector and matrix are

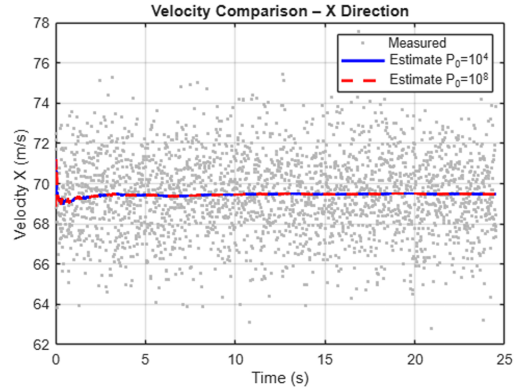
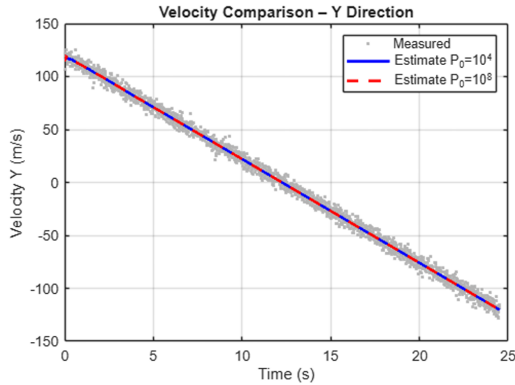
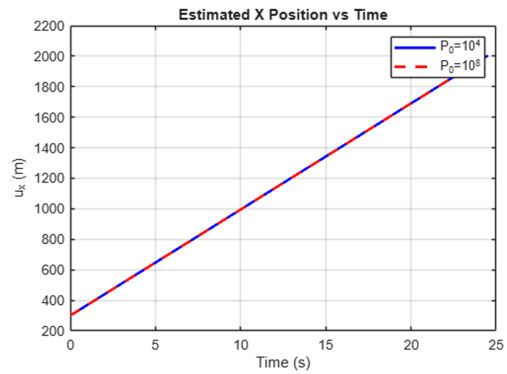
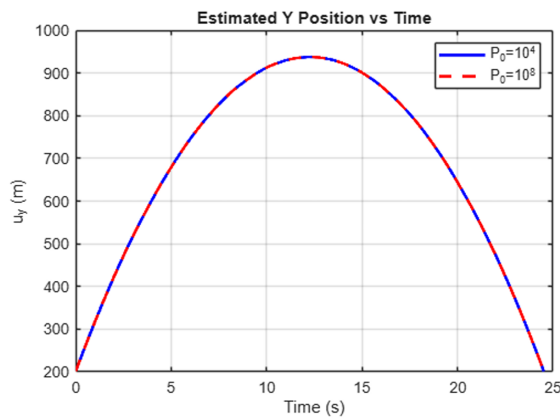
$$z = \begin{bmatrix} v_x \\ v_y \end{bmatrix}, \quad H = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad R = \text{diag}(4, 16).$$

Observations.

- Velocity states converge rapidly owing to direct measurement availability.
- Displacement states are reconstructed indirectly through the system dynamics.
- Position estimates exhibit slightly higher sensitivity to modelling assumptions compared to the displacement-measurement case.
- The Kalman filter successfully reconstructs the full projectile trajectory using only velocity measurements and physics-based dynamics.



(a) Trajectory using velocity measurements

(b) Velocity comparison — x -direction(a) Velocity comparison — y -direction(b) Estimated x -positionFigure 6: Estimated y -position

2 Question 2: Two-Story Structural System

The structural dynamics are governed by

$$M\ddot{y} + C\dot{y} + Ky = -M \mathbf{1} \ddot{u}_g,$$

with state vector $X = [y_1, y_2, \dot{y}_1, \dot{y}_2]^T$. The continuous-time state-space model is

$$\dot{X} = \begin{bmatrix} 0 & I \\ -M^{-1}K & -M^{-1}C \end{bmatrix} X + \begin{bmatrix} 0 \\ -\mathbf{1} \end{bmatrix} \ddot{u}_g.$$

The system was discretized using the matrix exponential. Ground motion sampled at 50 Hz was interpolated to match the measurement sampling rate of 200 Hz.

2.1 Q2(a): Acceleration-Only Measurements

Absolute accelerations $(\ddot{y}_i + \ddot{u}_g)$ are measured. The corresponding measurement matrix is

$$H = \begin{bmatrix} -M^{-1}K & -M^{-1}C \end{bmatrix}, \quad R = \text{diag}(0.105, 0.162).$$

Observations.

- Displacements and velocities are estimated entirely indirectly from acceleration observations.
- The estimation quality is more sensitive to modelling errors and measurement noise than in the displacement-measurement case.
- With $Q = 0$, the filter relies heavily on model predictions; introducing a small process noise covariance improves stability and smoothness.

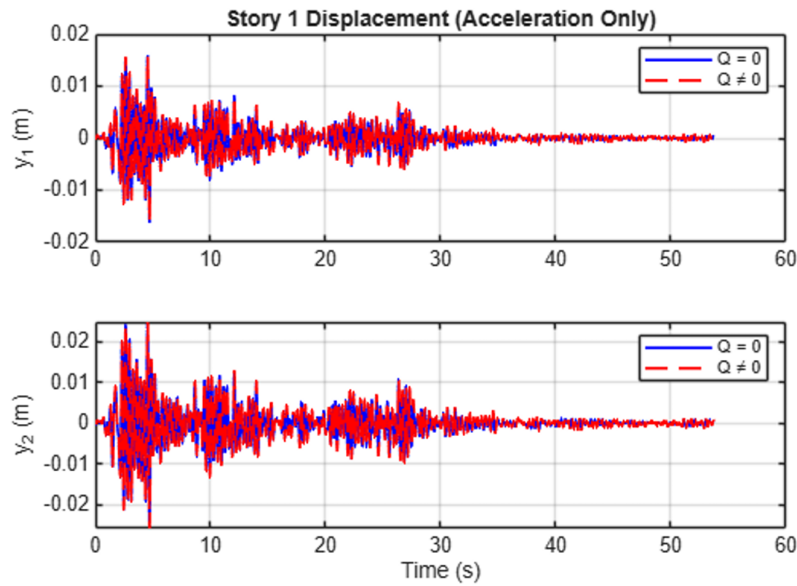


Figure 7: Estimated displacement response (acceleration-only measurements)

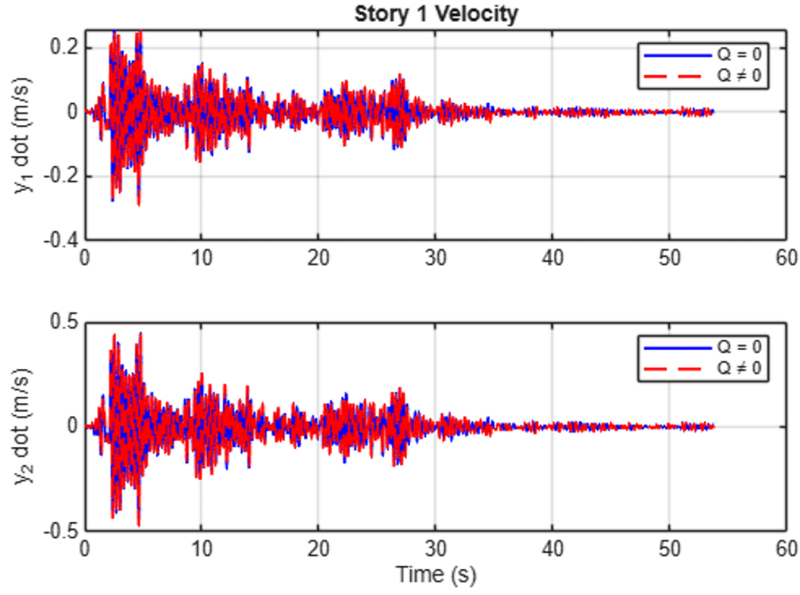


Figure 8: Estimated velocity response

2.2 Q2(b): Displacement-Only Measurements

The measurement vector and matrix are

$$z = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, \quad H = \begin{bmatrix} I & 0 \end{bmatrix}, \quad R = \text{diag}(2.642 \times 10^{-4}, 4.251 \times 10^{-4}).$$

Observations.

- Displacement states are directly corrected at each measurement step.
- Velocity states are estimated indirectly through state coupling in the dynamic model.
- Filter behaviour is smoother and more stable than the acceleration-only case.
- Sensitivity to modelling uncertainty is substantially reduced.

2.3 Comparison Between Measurement Types

Table 1: Comparison of acceleration-only vs. displacement-only estimation

Aspect	Acceleration Only	Displacement Only
Directly measured states	None	Displacements
Sensitivity to model error	Higher	Lower
Estimation stability	Moderate	Better
Need for process noise	More critical	Less critical

Acceleration-only estimation depends strongly on model accuracy because the measurements relate to second derivatives of the displacement states. In contrast,

direct displacement measurements provide immediate correction of the primary structural states, yielding more stable and reliable estimation performance overall.

2.4 Effect of Process Noise

Setting $Q = 0$ assumes a perfect model and may reduce the filter's ability to adapt to unmodelled dynamics or external disturbances. Introducing a small, appropriately tuned process noise covariance improves robustness and produces smoother, more physically realistic state tracking, at the cost of a modest increase in steady-state estimation variance.

3 Conclusion

The Kalman filter successfully tracked both projectile motion and the two-story structural response under earthquake excitation. The key findings are summarised as follows:

1. A larger initial error covariance (P_0) accelerates initial convergence by allowing the filter to weight early measurements more heavily.
2. The choice of measurement type (displacement vs. velocity vs. acceleration) strongly influences estimation behaviour, accuracy, and sensitivity to modelling errors.
3. Displacement measurements provide the most stable and direct estimation of structural states.
4. A small, non-zero process noise covariance Q improves robustness against model uncertainty and unmodelled dynamics.

This study demonstrates the effectiveness of Kalman filtering as a state estimation framework for structural health monitoring applications, covering both simple kinematic systems and multi-degree-of-freedom structural dynamics.